

Personality-assisted mood modeling with historical reviews for sentiment classification

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ABSTRACT

Review sentiment classification aims to predict user sentiment for given user-generated review. Most of the existing methods enhance their sentiment classifiers by incorporating user information. However, those methods normally ignore user's mood, which could influence her/his sentiment expression. Actually, users in a certain mood tend to express mood-congruent sentiment. Furthermore, the related studies may not fully utilize user's personality to model her/his mood. In this paper, we are motivated to propose a Personality-Assisted Mood for Sentiment Classification (PAMSC) model to classify user sentiment. Concretely, we first adopt the target review with global user preference to prejudge user sentiment. Meanwhile, we model the personalized mood impact on her/his sentiment expression according to the corresponding historical reviews. We finally obtain the prediction result by utilizing the personalized mood impact to adjust the prejudged sentiment distribution. Particularly, personality plays two major roles in the process of mood modeling. One is to analyze the personalized duration of user's mood, and the other is to model the personalized attention toward mood-congruent information. The experimental results demonstrate that our PAMSC model not only achieves the highest classification accuracy than the related models on three real-world datasets but also has stronger interpretability for the prediction process.

1. Introduction

Review sentiment classification is a fundamental problem in the field of sentiment analysis and opinion mining [30]. It aims to infer user sentiment according to a given user-generated review. The inferred user sentiment could play essential roles in many fields, including recommender system [37], depression detection [44], and stock market [11].

The related methods normally regard sentiment classification as a special case of text classification [29]. They generally utilize machine learning or deep learning to build sentiment classifiers based on text information [46]. Recently, some researchers have attempted to enhance their sentiment classifiers by incorporating user information (e.g., global user preference [14], and dynamic user interest [15]), because user information has a significant influence on user's sentiment expression [5].

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However, the majority of the existing studies normally do not consider the user's mood hidden in historical reviews. Concretely, mood is a long-term diffuse affective state that typically has positive and negative valence [3]. Previous psychological findings have shown that a user's current mood predisposes her or him to express sentiment consistent with the current mood [9]. For example, users who have a positive/negative mood are prone to express positive/negative sentiment. Furthermore, few of them make full use of the user's personality when analyzing the user's mood. In fact, as an affect-processing system, the user's personality not only affects the duration of the user's mood but also influences the user's attention toward mood-congruent information under different moods [20,34].

In this work, we are motivated to propose a Personality-Assisted Mood for Sentiment Classification (PAMSC) model, where the idea is inspired by the cognitive mechanism of mood-congruent attention bias [1]. Specifically, this cognitive mechanism makes users give more attention to mood-congruent information, which leads to the expression of mood-congruent sentiment [19]. For example, users in a negative mood tend to focus on negative information and express negative sentiment. While for users who have a positive mood, s/he is prone to concentrate on positive information and express positive sentiment. Concretely, for our PAMSC model, we first integrate the target review with global user preference to prejudge user sentiment. Simultaneously, according to the corresponding historical reviews of the target review, we model the personalized impact of the user's current mood, which is beneficial for increasing the attention toward mood-congruent sentiment classes. Finally, we adjust the prejudged sentiment distribution with the personalized mood impact to obtain the final classification result. In particular, the role of personality in our PAMSC model is twofold: (1) To analyze the duration of the user's personalized mood for extracting the personalized mood from historical reviews; (2) To analyze the user's personalized attention toward mood-congruent information for modeling the personalized mood impact. The experimental results demonstrate that our PAMSC model significantly outperforms the related models on three real-world datasets in terms of classification accuracy. In addition, compared with the related models that do not use the personalized mood impact, our PAMSC model accomplishes the sentiment classification task in a more interpretable way.

In summary, the main contributions of our work are as follows:

(1) We have proposed a personality-assisted mood for sentiment classification model to classify user sentiment. Compared with other related models, our PAMSC model fully utilizes user's historical reviews to mine the corresponding user's mood, thus precisely analyzing user sentiment with the personalized mood impact.

(2) We have utilized the user's personality to model her or his mood in a more personalized manner, which provides our PAMSC model with more detailed information about the user's mood and enables our PAMSC model to accurately complete the sentiment classification task.

(3) We have conducted experiments on three real-world datasets to verify the effectiveness of our PAMSC model. The experimental results demonstrate that our PAMSC model achieves better classification accuracy and interpretation than the related models on three real-world datasets.

In the following sections, we first introduce related work regarding review sentiment classification in Section 2. We then describe our proposed PAMSC model in Section 3. We further present the details of our experimental design and analyze the experimental results in Section 4. After that, we summarize our experimental findings and discuss the limitations of our work in Section 5. Finally, we conclude the paper and indicate some future directions in Section 6.

2. Related work

Most of the related studies treated the sentiment classification task as a special case of text classification problem [29]. They normally took annotated sentiment polarities or ratings as sentiment classes (e.g., 1–5 stars represent strongly negative, weakly negative, neutral, weakly positive, and strongly positive sentiment classes, respectively) and adopted machine learning algorithms to build sentiment classifiers with hand-craft text features, such as text topics [10]. However, the performance of the sentiment classifiers largely depends on feature engineering, which is time-consuming, labor-intensive, and requires sufficient expert knowledge.

The emergence of Long-Short Term Memory (LSTM) allows the researchers to no longer focus on feature engineering due to its ability to capture the sequential information. Specifically, Tai et al. [40] introduced tree-structured networks into LSTM for better semantic composition. Xu et al. [45] divided LSTM cells into several groups with different forgetting rates for capturing local and global semantic features. In addition, motivated by Neural Machine Translation (NMT) [17], attention mechanism is introduced into sentiment classification because it can help the model focus on the important words in a review and generate better review representation. For instance, Zhang et al. [46] used multiple view vectors to map the attention from different perspectives for learning review representation. Zhu et al. [49] proposed a dual-channel label-guided attention network to learn text representation via two different channels.

In recent years, Pretrained Language Models (PLMs) (e.g., Robustly Optimized BERT Pretraining Approach (RoBERTa) [23]) have been widely used in sentiment classification task for analyzing text information and generating effective text representations [39]. For example, Ke et al. [18] introduced word-level linguistic knowledge into PLMs to construct knowledge-aware language representations. Zhao et al. [48] design two sentiment-aware auxiliary tasks to integrate sentiment knowledge in the fine-tuning stage for enhancing PLMs. However, these models usually have high computational costs and time complexity because of a large number of parameters.

In addition, considering that user information could significantly influence user's sentiment expression, some researchers have endeavored to incorporate user information into their models to improve the performance of sentiment classification. Specifically, Chen et al. [5] took into account global user preference at both word level and semantic level by user attention to generate better review representations. Lin et al. [22] utilized user's personality to extract personalized sentiment features to train sentiment classi-

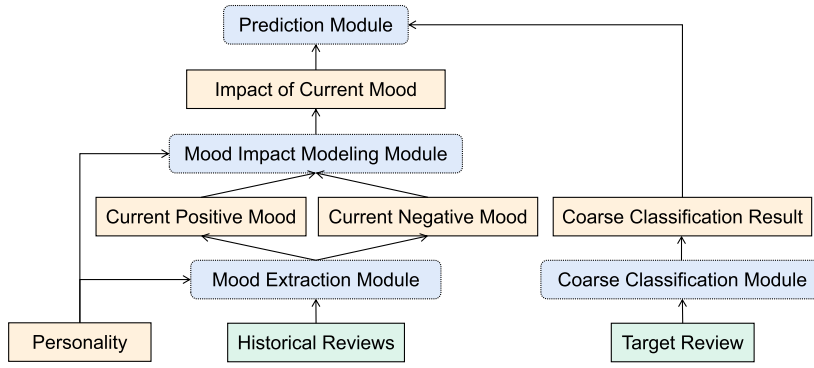


Fig. 1. The whole architecture of PAMSC model.

fier. Long et al. [24] utilized memory network [38] to capture user information for sentiment prediction. Lyu et al. [26] explicitly stored representations of reviews written by the same user and forced the model to memorize all reviews for one particular user. Zhang et al. [47] modeled the interactive relationship between user information and text information to improve sentiment classification performance. Mirehghallah et al. [28] added a fixed, user-specific non-trainable string to each user's review for capturing user information. Additionally, some related studies attempted to mine dynamic user information by using user's historical reviews. Jia et al. [15] designed two attentive memory networks to utilize user's historical reviews to model the temporal dynamics in user's interest and temperament.

However, the above-mentioned methods failed to achieve satisfactory performance on sentiment classification task. As for the possible reason, although some methods attempt to mine dynamic user information (e.g., dynamic user interest [15]) from historical reviews to understand user sentiment, they normally fail to consider user's mood hidden in historical reviews. As a diffuse and enduring affective state, a user's mood could make her or him express mood-congruent sentiment [9,19]. Furthermore, most of the related studies do not fully utilize user's personality to model the personalized mood impact. Personality is defined as an affect processing system that describes the relatively stable pattern of humans in terms of their behaviors, cognition, and desire over time and space [35]. Psychologists have found that the duration of mood is influenced by one's personality [20]. For instance, neurotic users tend to go into a negative mood easily and change their mood frequently. While for agreeable users, s/he is prone to go to a positive mood more often. In addition, personality also influences the attention deployment of users toward mood-congruent information under different moods [34]. For example, extroverted users tend to focus on positive information in a positive mood than in a negative mood. In contrast, neurotic users are prone to pay more attention to negative information in a negative mood than in a positive mood. Although user's mood and personality have been widely applied in various fields (e.g., recommender system [7]), few researchers have considered them when dealing with sentiment classification task. Hence, we are interested in modeling the user's mood with the corresponding historical reviews and simulating the impact of the user's mood for assisting sentiment classification. We also would like to see the role of personality in modeling the personalized mood impact.

3. Methodology

Fig. 1 depicts the overall architecture of our PAMSC model, which includes four modules. Note that, the personality data used in our PAMSC model is extracted from user-generated reviews based on the findings of linguistic psychology. Concretely, in *Mood Extraction Module*, PAMSC model analyzes the duration of the user's current mood with her or his personality, and calculates the current mood based on the corresponding historical reviews. After that, in *Mood Impact Modeling Module*, PAMSC model integrates the user's current mood with her or his personality to model the personalized impact of the user's current mood. Meanwhile, PAMSC model also adopts global user preference to analyze the target review and prejudge user sentiment in *Coarse Classification Module*. Finally, in *Prediction Module*, PAMSC model uses the personalized impact of the user's current mood to adjust the sentiment distribution based on the coarse classification result and obtains the final classification result. We will give the formalizations first and then describe the process of personality extraction. Finally, we will introduce the details of our PAMSC model in a module-by-module way.

3.1. Formalizations

For M -class sentiment classification task, a review r consists of l_r words as $\{w_1^r, w_2^r, \dots, w_{l_r}^r\}$. Then, the sentiment classification task is defined as: Given a sequence $\mathbf{R}^u = \{r_1, r_2, \dots, r_n\}$ of n reviews in time order posted by user u , we want to predict the i -th ($1 \leq i \leq n$) review's sentiment class. In this scenario, we call r_i as the target review and other reviews posted before r_i as its corresponding historical reviews. In addition, $\mathbf{E}^u$ and $\mathbf{G}^u$ denote the personality and global preference of u , respectively.

Table 1
The strongly correlated LIWC2015 word categories of Big-Five personality traits.

Trait	Level	LIWC2015 word categories
O	High	Insight, Death, Percept, Cogproc, Hear
	Low	Netspeak, Family, Posemo, Reward, Affect
C	High	Drives, Achieve, Time, Posemo, Work
	Low	Anger, Negemo, Sexual, Swear, Body
E	High	Posemo, Affiliation, Netspeak, Reward, Friends
	Low	Death, Cogproc, Tentat, Insight, Differ
A	High	Posemo, Drives, Affiliation, Time, Relativ
	Low	Anger, Swear, Negemo, Sexual, Death
N	High	Negemo, Anger, Sad, Anx, Death
	Low	Posemo, Affiliation, Reward, Achieve, Leisure

3.2. Personality extraction

The Big-Five personality model is one of the most well-known and authoritative personality models, which describes the user's personality based on five traits: *Openness to experience (O)*, *Conscientiousness (C)*, *Extraversion (E)*, *Agreeableness (A)*, and *Neuroticism (N)* [12]. Hence, we adopt Big-Five personality model to describe the user's personality. Specifically, we extract user's personality traits based on Linguistic Inquiry and Word Count 2015 (LIWC2015) [31] word categories, as it has been proven to be related to Big-Five personality [8]. The reason why we do not use the existing transformer-based personality prediction models [50] is that most of them are trained on the Essays dataset [32]. This might result in poor performance of these transformer-based personality prediction models on the datasets used in our work due to a significant time gap in data generation between the Essays dataset and the datasets we used. In contrast, LIWC2015 is a relatively new linguistic research achievement and has been proven effective by a large number of researchers [6]. Similar to [43], we pick the top five LIWC2015 word categories, which are strongly correlated with the corresponding personality trait and high/low level. Table 1 shows the chosen 50 word categories, 10 for each personality trait. We then calculate the Term Frequency-Inverse Document Frequency (TF-IDF) value of the words for the 50 word categories to denote the user's personality. Note that, the user's personality is calculated based on the user's all target reviews in the training set.

Concretely, for user u , we first calculate the Term Frequency (TF) for each LIWC2015 word category:

$$TF^u = [TF_1, TF_2, \dots, TF_{50}]$$

$$= [\frac{E_1}{L^{tf}}, \frac{E_2}{L^{tf}}, \dots, \frac{E_{50}}{L^{tf}}] \quad (1)$$

where E_i is the number of times that the word of the i -th LIWC2015 word category appears in user u 's target reviews, while L^{tf} is the total number of words in user u 's target reviews.

Then, for each LIWC2015 word category, we calculate the corresponding Inverse Document Frequency (IDF):

$$IDF^u = [IDF_1, IDF_2, \dots, IDF_{50}]$$

$$= [\log \frac{L^{idf}}{G_1 + 1}, \log \frac{L^{idf}}{G_2 + 1}, \dots, \log \frac{L^{idf}}{G_{50} + 1}] \quad (2)$$

where G_i is the number of user u 's target reviews that contain words of the i -th LIWC2015 word category, while L^{idf} is the total number of user u 's target reviews.

Finally, the Big-Five personality of user u is calculated as:

$$E^u = [TF_1 \times IDF_1, TF_2 \times IDF_2, \dots, TF_{50} \times IDF_{50}] \quad (3)$$

3.3. Mood extraction module

We model the user's current mood according to the proportion of positive words used in the corresponding historical reviews, as psychologists have found that users in a positive/negative mood are prone to use more positive/negative words when writing [27]. In addition, unlike emotion which is only experienced for an instant, mood can last for a long period [2]. Therefore, it is important to select historical reviews with appropriate time intervals to model the corresponding mood. Considering that the user's personality could affect the duration of her or his mood to a certain extent [16], we utilize the user's personality to select appropriate historical reviews and assess the duration of the current mood.

Specifically, we first set a sliding time window over the historical reviews to divide them into different blocks. To balance the timeliness of the current mood and the sufficiency of the data, the time window is set as 1 month and the number of slides is set as 6. Hence, we obtain 6 blocks of the historical reviews, and the i -th block contains the historical reviews posted in the i -th month before the target review. After that, we adopt VADER-Sentiment-Analysis [13] to count the number of positive words used in the historical reviews of each block. Formally, the number of positive words used in 6 blocks is defined as $S^p = \{s_1^p, s_2^p, \dots, s_6^p\}$, where s_i^p is the number of positive words used in the i -th block. Similarly, the number of negative words used in 6 blocks is defined as $S^n = \{s_1^n, s_2^n, \dots, s_6^n\}$. Then, the proportion of positive words used in 6 blocks is defined as $S^{pn} = \{s_1^{pn}, s_2^{pn}, \dots, s_6^{pn}\}$, where $s_i^{pn} = \frac{s_i^p}{s_i^p + s_i^n}$ is the

proportion of positive words used in the i -th block. Note that, $s_i^{pn} = 0.5$ when $s_i^p = s_i^n = 0$. Furthermore, we adopt the user's personality to analyze the importance of the 6 blocks, and the current positive mood $\mathbf{D}^{pos} \in [0, 1]$ is a weighted summation of s_i^{pn} as:

$$\mathbf{D}^{pos} = \sum_{i=1}^6 \alpha_i s_i^{pn} \quad (4)$$

where α_i is the attention weight of s_i^{pn} , which measures the importance of the i -th block. In addition, $\alpha = \{\alpha_1, \alpha_2, \dots, \alpha_6\}$ is calculated by:

$$\alpha = softmax(tanh(\mathbf{W}^{mee} \mathbf{E}^u + \mathbf{W}^{mes} \mathbf{S}^{pn} + \mathbf{b}^{me})) \quad (5)$$

where $\mathbf{W}^{mee}$ and $\mathbf{W}^{mes}$ are weight matrices. $\mathbf{b}^{me}$ is weight vector.

Moreover, the current negative mood $\mathbf{D}^{neg} \in [0, 1]$ is defined as:

$$\mathbf{D}^{neg} = 1 - \mathbf{D}^{pos} \quad (6)$$

3.4. Mood impact modeling module

We model the personalized impact of the current mood according to mood-congruent attentional bias. Concretely, the mood impact is used to increase the attention toward the mood-congruent sentiment classes. The higher the consistency with the current mood, the more the attention increases. Specifically, to model the impact of the current positive mood, we first introduce a basic positive-based attention vector $\mathbf{Z}^{pb} = [\frac{1}{M}, \frac{2}{M}, \dots, \frac{M-1}{M}, \frac{M}{M}] \in \mathbb{R}^M$ to ensure that larger sentiment classes (with more positive sentiment) receive more attention under a positive mood. Note that, M is the number of sentiment classes and $\mathbf{Z}^{pb}$ is non-trainable. Considering that the fixed $\mathbf{Z}^{pb}$ could not represent the personalized attention toward mood-congruent information of different users, we further adopt the user's personality to adjust the attention distribution of $\mathbf{Z}^{pb}$ to obtain the personalized positive-based attention vector $\mathbf{Z}^{pp}$, as the user's personality could influence the user's attention toward mood-congruent information under different moods [34]. Finally, we obtain the current positive mood's impact $\mathbf{A}^{pos} \in \mathbb{R}^M$ by incorporating the extracted current positive mood. Formally, $\mathbf{A}^{pos}$ is calculated as follows (Eqs. (7)-(9)), and each dimension of $\mathbf{A}^{pos}$ represents the attention toward the corresponding sentiment class under the current positive mood.

$$\mathbf{A}^{pos} = softmax(\mathbf{D}^{pos} * tanh(\mathbf{Z}^{pp})) \quad (7)$$

$$\mathbf{Z}_i^{pp} = \mathbf{Z}_i^{pb} + z^{pe} \quad (i \text{ from } 1 \text{ to } M) \quad (8)$$

$$z^{pe} = relu(\mathbf{W}^{mcp} \mathbf{E}^u) \quad (9)$$

where $Relu(x) = max(0, x)$ is activation function. $\mathbf{W}^{mcp}$ is weight matrice.

Similarly, the current negative mood's impact $\mathbf{A}^{neg} \in \mathbb{R}^M$ is calculated by:

$$\mathbf{A}^{neg} = softmax(\mathbf{D}^{neg} * tanh(\mathbf{Z}^{np})) \quad (10)$$

$$\mathbf{Z}_i^{np} = \mathbf{Z}_i^{nb} + z^{ne} \quad (i \text{ from } 1 \text{ to } M) \quad (11)$$

$$z^{ne} = relu(\mathbf{W}^{mcn} \mathbf{E}^u) \quad (12)$$

where $\mathbf{Z}^{nb} = [\frac{M}{M}, \frac{M-1}{M}, \dots, \frac{2}{M}, \frac{1}{M}] \in \mathbb{R}^M$ is the basic negative-based attention vector of negative mood and $\mathbf{Z}^{nb}$ is non-trainable. We also use it to ensure that the smaller sentiment classes (with more negative sentiment) could receive more attention under a negative mood. In addition, $\mathbf{W}^{mcn}$ is weight matrice.

Finally, the current mood's impact $\mathbf{A}^{pn} \in \mathbb{R}^M$ is a summation of the impact of the current positive and negative moods:

$$\mathbf{A}^{pn} = \mathbf{A}^{pos} + \mathbf{A}^{neg} \quad (13)$$

3.5. Coarse classification module

In this module (see Fig. 2), we use Bi-directional LSTM (Bi-LSTM) with attention mechanism to generate the representation of the target review, and then utilize the review representation to obtain coarse classification result. Specifically, we first adopt a pretrained RoBERTa to generate the contextualized representation of each word. We further utilize Bi-LSTM to generate the hidden representation of each word, as Bi-LSTM is good at capturing sequential information of text. Formally, the hidden representation $\mathbf{h}_i$ of the i -th word is obtained by:

$$\mathbf{h}_i = [\bar{\mathbf{h}}_i, \mathbf{h}_i], \quad \bar{\mathbf{h}}_i = \overrightarrow{LSTM}(\bar{\mathbf{h}}_{i-1}, \mathbf{e}_i), \quad \mathbf{h}_i = \overleftarrow{LSTM}(\mathbf{h}_{i+1}, \mathbf{e}_i) \quad (14)$$

where $\mathbf{e}_i$ denotes the contextualized representation of the i -th word. In addition, $\bar{\mathbf{h}}_i$ and $\mathbf{h}_i$ represent the forward and backward hidden representations respectively.

Inspired by [47], we design two encoders (i.e., Text-to-Text Encoder and User-to-Text Encoder) to encode the target review. For the text-to-text encoder, we adopt self-attention [42] to obtain the text-to-text based review representation $\mathbf{r}''$. Formally:

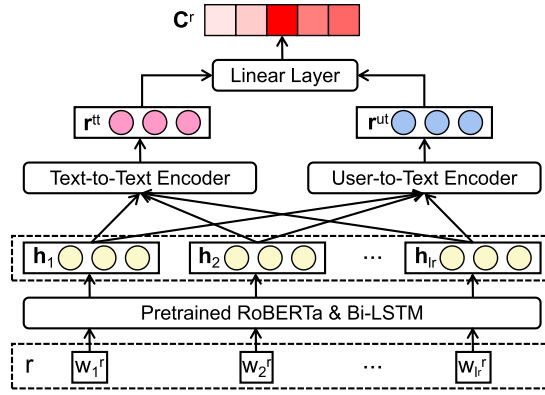


Fig. 2. The architecture of coarse classification module.

$$\mathbf{r}^{tt} = \sum_{i=1}^{l_r} \beta_i \mathbf{h}_i \quad (15)$$

where β_i is the attention weight of $\mathbf{h}_i$, which measures the importance of the i -th word from the perspective of the text-to-text encoder. The attention weight β_i for each hidden state is defined as:

$$\beta_i = \frac{\exp(f^{tt}(\mathbf{h}_i))}{\sum_{j=1}^{l_r} \exp(f^{tt}(\mathbf{h}_j))} \quad (16)$$

$$f^{tt}(\mathbf{h}_i) = \mathbf{v}^{ttT} \tanh(\mathbf{W}^{tt} \mathbf{h}_i + \mathbf{b}^{tt}) \quad (17)$$

where f^{tt} is a score function that measures the importance of words for composing the text-to-text based review representation. $\mathbf{W}^{tt}$ is weight matrix. $\mathbf{v}^{tt}$ is weight vector and $\mathbf{v}^{ttT}$ represents its transpose.

For the user-to-text encoder, considering that different users have different word and class preferences [14], we adopt global user preference as user attention [5] to help our model focus on the specific words that have greater contributions to the meaning of the review. Then, the review representation is formed by aggregating the representations of those specific words. Formally, the user-to-text based review representation $\mathbf{r}^{ut}$ is a weighted summation of hidden states as:

$$\mathbf{r}^{ut} = \sum_{i=1}^{l_r} \gamma_i \mathbf{h}_i \quad (18)$$

where γ_i is the attention weight of $\mathbf{h}_i$, which measures the importance of the i -th word from the view of the user-to-text encoder. The attention weight γ_i for each hidden state is defined as:

$$\gamma_i = \frac{\exp(f^{ut}(\mathbf{h}_i, \mathbf{G}^u))}{\sum_{j=1}^{l_r} \exp(f^{ut}(\mathbf{h}_j, \mathbf{G}^u))} \quad (19)$$

$$f^{ut}(\mathbf{h}_i, \mathbf{G}^u) = \mathbf{v}^{utT} \tanh(\mathbf{W}^{uth} \mathbf{h}_i + \mathbf{W}^{utg} \mathbf{G}^u + \mathbf{b}^{ut}) \quad (20)$$

where f^{ut} is a score function that measures the importance of words for composing the user-to-text based review representation. $\mathbf{W}^{uth}$ and $\mathbf{W}^{utg}$ are weight matrices. $\mathbf{v}^{ut}$ is weight vector and $\mathbf{v}^{utT}$ represents its transpose.

Finally, we adopt a linear layer to project $\mathbf{r}^{tt}$ and $\mathbf{r}^{ut}$ into the target space of M classes and obtain the coarse classification result $\mathbf{C}^r \in \mathbb{R}^M$ (see Eq. (21)). Each dimension of $\mathbf{C}^r$ denotes the probability that the corresponding sentiment class belongs to the real sentiment class of the target review.

$$\mathbf{C}^r = \mathbf{W}^c [\mathbf{r}^{tt}; \mathbf{r}^{ut}] + \mathbf{b}^c \quad (21)$$

where $\mathbf{W}^c$ is weight matrix and $\mathbf{b}^c$ is weight vector.

3.6. Prediction module

We adopt the mood impact $\mathbf{A}^{pn}$ to adjust the user's sentiment distribution based on the coarse classification result $\mathbf{C}^r$ to obtain the final result $\mathbf{F}^r$ and use a softmax layer to project $\mathbf{F}^r$ into the user sentiment distribution of M classes. Specifically,

$$\mathbf{F}^r = \mathbf{C}^r + \mathbf{A}^{pn} \quad (22)$$

$$\hat{y} = \text{softmax}(\mathbf{F}^r) \quad (23)$$

where $\hat{y}$ is the prediction sentiment distribution of $\mathbf{F}^r$.

Table 2
Statistics of Yelp2018, Yelp2019, and Yelp2020 datasets.

Dataset	Yelp2018	Yelp2019	Yelp2020
#sentiment classes	5	5	5
#users	2,773	2,759	1,571
#target reviews	55,062	54,594	28,939
#words per target review	154.48	155.19	147.56

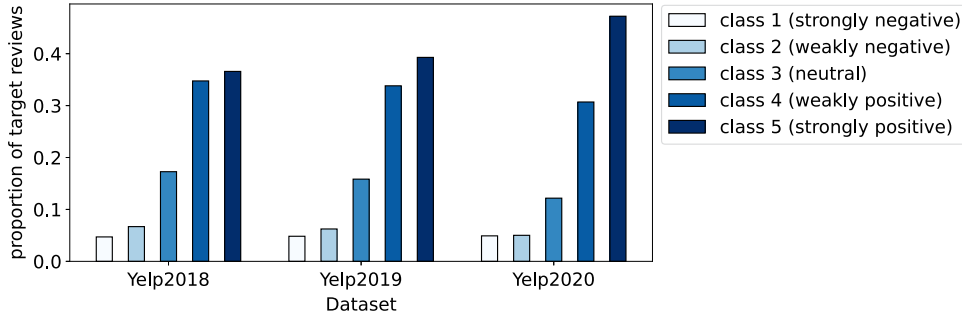


Fig. 3. The class distribution of Yelp2018, Yelp2019, and Yelp2020 datasets.

In our model, the loss function is defined as the cross-entropy error between the ground truth distribution y and $\hat{y}$:

$$loss = - \sum_{r \in \mathbf{R}} \sum_{m=1}^M y_m(r) \cdot \log(\hat{y}_m(r)) \quad (24)$$

where $\mathbf{R}$ denotes the training set. In addition, y_m is the gold probability of sentiment class m with the ground truth being 1 and other being 0.

4. Experiments

In this section, we conduct experiments on three real-world datasets to validate the effectiveness of our PAMSC model. Specifically, we aim to answer the following Research Questions (RQs):

RQ1: Does our proposed PAMSC model outperform the related baseline models? (see Section 4.5)

RQ2: Whether the current positive and negative moods affect users' sentiment expression? (see Section 4.6)

RQ3: Is personality effective for analyzing the personalized duration of the current mood? (see Section 4.7)

RQ4: Is personality effective for analyzing the personalized attention of users toward mood-congruent information? (see Section 4.7)

4.1. Dataset

We build three datasets (i.e., Yelp2018, Yelp2019, and Yelp2020) based on the original Yelp dataset.¹ The reason why we choose the Yelp dataset is that it provides the time data accurate to a second (i.e., the format of the time is %Y-%m-%d %H:%M:%S), whereas the time data provided by Amazon and others are normally only accurate to a day (i.e., the format of the time is %Y-%m-%d), which does not meet our need to sort the reviews chronologically. Concretely, we first extract the reviews within the specified year (e.g., 2018, 2019, and 2020) from the original Yelp dataset. For each user, we sort her or his reviews in time order. For each review, we label it to be a target review and regard all reviews posted before it as its historical reviews. To fully study user information, we retain the target reviews with at least 6 months of history and exclude users with less than 5 target reviews. The statistics of the three generated datasets are summarized in Table 2. As shown in Table 2, the scale of Yelp2020 is smaller than that of Yelp2018 and Yelp2019, which may be caused by the impact of the COVID-19 pandemic. Finally, according to the time order, we split all target reviews of each user into training, development, and test sets in the proportion of 8:1:1. In addition, similar to [41], we adopt human-labeled review ratings as the gold standard sentiment classes, and the class distribution of the three datasets is shown in Fig. 3. It can be seen that the class distribution of the three datasets is basically consistent.

¹ <http://www.yelp.com/dataset>.

4.2. Evaluation metrics

We use *Accuracy* (short as *Acc.*, the higher the better) and *Mean Absolute Error* (*MAE*, the lower the better) as the evaluation metrics. *Accuracy* is used to measure the overall sentiment classification performance, while *MAE* is used to measure the divergences between predicted sentiment classes and ground truth classes. Formally, in our case:

$$Accuracy = \frac{T}{N} \quad (25)$$

$$MAE = \frac{\sum_{i=1}^N |gd_i - pr_i|}{N} \quad (26)$$

where T is the number of predicted sentiment classes that are similar to gold sentiment classes, and N is the number of target reviews. In addition, gd_i and pr_i denote the gold sentiment class and predicted sentiment class, respectively.

4.3. Baselines

Based on our literature research, we compare our PAMSC model with several related models used for review sentiment classification. These models can be divided into two categories: *models without/with user information*. Note that, to the best of our knowledge, PbSC [22] is the only model that integrates user personality into the sentiment classification task. However, on the one hand, due to the lack of detailed information about the usage of user personality, we cannot reproduce PbSC. On the other hand, PbSC only adopts machine learning algorithms (e.g., decision tree) as its classifier, which makes it unable to be fairly compared with other deep learning-based sentiment classification models (e.g., PAMSC and IAAN). Hence, we do not choose PbSC as a comparison method.

4.3.1. Models without user information

- **RoBERTa (FiT)** [23] fine-tunes pretrained RoBERTa and sends the representation of [CLS] to a linear layer for classifying user sentiment.
- **NSC+LA** [5] introduces local semantic attention into LSTM to classify user sentiment.
- **NSC+LA (Bi-LSTM)** implements NSC+LA with Bi-LSTM.
- **MVA** [46] obtains review representations from different perspectives by multiview attention and regularization terms.

4.3.2. Models with user information

- **NSC+UA** [5] integrates user attention (i.e., global user preference) with LSTM for sentiment classification.
- **NSC+UA (Bi-LSTM)** implements NSC+UA with Bi-LSTM.
- **SentiMem** [15] captures the user's dynamic interest and temperament from the historical reviews through two memory networks (i.e., interests memory network and temperaments memory network). The temporal dynamics of the user's interest and temperament mined are used to identify user sentiment in the target review.
- **IAAN** [47] captures the interaction between user information, product information, and text information through bilinear interactive attention for predicting user sentiment.
- **UserIdentifier** [28] prepends a fixed, user-specific, and non-trainable string to each user's review for capturing user information and classifying user sentiment.

Chen et al. [5] do not implement NSC+LA and NSC+UA with Bi-LSTM. Considering that MVA, IAAN, and PAMSC use Bi-LSTM, we implement them (i.e., NSC+LA (Bi-LSTM) and NSC+UA (Bi-LSTM)) in our experiments to make the experimental results more convincing.

4.4. Implementation details

Since our PDMSR model does not use product information, we only keep the processing of text information and user information in IAAN. In addition, considering that NSC+LA, NSC+LA (Bi-LSTM), NSC+UA, NSC+UA (Bi-LSTM), and IAAN model the reviews from the word-level to the sentence-level through the LSTM-based hierarchical structure, we only retain the part of word-level review modeling because the reviews in our three real-world datasets are not cut into sentences. Moreover, we replace the different word embeddings used in the baseline models with the fixed contextualized word representations generated by the pretrained RoBERTa.

Furthermore, we use the same experimental setting for all models. Concretely, to speed up training, we limit each review has no more than 128 words (padding when the review length is less than 128, truncation when the review length is greater than 128). In addition, we utilize AdamW [25] as these models' optimizers for updating parameters when training. Meanwhile, we employ the early stopping strategy with a patience of 4 to avoid overfitting. Besides, we select the best parameters based on the performance of the development set and evaluate the parameters on the test set. Specifically, the dimension of the hidden state in RNN/LSTM/GRU cell is set to 128, the batch size is chosen from {16, 32, 64}, and the learning rate is selected from {1e-3, 1e-4, 1e-5}. Moreover, the number of view vectors of MVA is set to 3, the number of historical reviews of SentiMem is set to 5, and the length of the user identifier of UserIdentifier is set to 10. Moreover, for our PAMSC model, G^u (i.e., the global preference of u) is randomly initialized from a uniform distribution $U(-0.01, 0.01)$ and its dimension is set to 128.

Table 3

The experimental results for review sentiment classification. * denotes the statistical significance for p -value<0.01 compared to the best baseline, the boldface indicates the best model results of the dataset, and the underline indicates the second best model result of the dataset.

Model	Year	Yelp2018		Yelp2019		Yelp2020	
		Accuracy	MAE	Accuracy	MAE	Accuracy	MAE
Model without user information							
RoBerta (FiT)	2019	63.0%	0.414	63.5%	0.410	67.1%	0.370
NSC+LA	2016	62.7%	0.419	64.0%	0.413	67.3%	0.380
NSC+LA (Bi-LSTM)	2016	62.6%	0.421	63.0%	0.419	66.8%	0.384
MVA	2021	61.8%	0.425	62.6%	0.422	66.2%	0.388
Model with user information							
NSC+UA	2016	65.1%	0.396	<u>66.6%</u>	0.381	69.2%	0.357
NSC+UA (Bi-LSTM)	2016	64.6%	0.404	66.3%	0.387	69.5%	0.357
SentiMem	2020	64.0%	0.415	66.0%	0.387	69.6%	<u>0.344</u>
IAAN	2021	65.3%	0.391	66.3%	0.390	<u>69.7%</u>	<u>0.344</u>
UserIdentifier	2022	<u>65.6%</u>	<u>0.378</u>	66.0%	<u>0.379</u>	69.4%	0.346
Ours: PAMSC	2023	66.6%*	0.377	67.0%*	0.374*	70.4%*	0.341

4.5. Model comparisons

To make a fair comparison between our proposed PAMSC model and other baseline models on the three real-world datasets, we implement all of the baseline models and report the mean results of five runs. The experimental results are shown in Table 3, and we adopt permutation test [33] for the significance test because the data is not normal distribution.

Firstly, to answer **RQ1**, we observe from Table 3 that our PAMSC model significantly outperforms other baseline models w.r.t. classification accuracy ($p < 0.01$ via permutation test) in all three datasets regardless of their scale (i.e., Yelp2020 dataset is of small scale, while Yelp2018 and Yelp2019 datasets are of relatively larger scale). For example, our PAMSC model increases its classification accuracy from 69.4% to 70.4% on Yelp2020 dataset when compared to UserIdentifier. As for the reason, based on the cognitive mechanism of mood-congruent attention bias, the personality-assisted mood we designed enables our PAMSC model to pay more attention to the sentiment classes that are consistent with the current mood, which is beneficial for classifying user sentiment in a fine-grained manner from the user's view.

Furthermore, compared with the baseline models that only utilize global or dynamic user information, our PAMSC model achieves better performance by combining global user information (e.g., user's preference and personality) and dynamic user information (user's current mood). For instance, relative to IAAN (i.e., the model that only uses global user information), our PAMSC model decreases its MAE on Yelp2019 dataset from 0.390 to 0.374. Similarly, our PAMSC model decreases its MAE from 0.415 to 0.377 on Yelp2018 dataset when compared to SentiMem (i.e., the model that only utilizes dynamic user information). This demonstrates that both global and dynamic user information play important roles in sentiment classification task.

Another interesting observation is that our PAMSC model performs much better than SentiMem, which is the only model using historical reviews among all baseline models. For example, compared with SentiMem, our PAMSC model increases its classification accuracy from 66.0% to 67.0% on Yelp2019 dataset. The possible reason for the poor performance of SentiMem is that it cannot capture the temporal dynamics of user's interest in our datasets. Concretely, our datasets (i.e., Yelp2018, Yelp2019, and Yelp2020) contain reviews written by users for different categories of products. Hence, it is difficult for SentiMem to mine appropriate user's interests from historical reviews to classify the user's sentiment of the corresponding target review. However, our PAMSC model only utilizes the polarity (i.e., positive and negative) of words to analyze the current mood from historical reviews. As the current mood is product-independent, our PAMSC model is suitable for a wider range of scenarios (e.g., the scenario with no product information). Another possible reason is that the user's mood might have a greater impact on her or his sentiment expression than the user's dynamic interest and temperament. For example, when a user is in a negative mood, even if s/he gets a product that suits her or his interest, s/he may still be depressed and express negative sentiment.

4.6. Effect of mood

We design PAMSC-M to evaluate the effect of user's mood and answer **RQ2**. Specifically, PAMSC-M removes the mood extraction module. It calculates A^{pos} and A^{neg} by modifying Eq. (7) and Eq. (10):

$$A^{pos} = \text{softmax}(\tanh(Z^{pp})) \quad (27)$$

$$A^{neg} = \text{softmax}(\tanh(Z^{np})) \quad (28)$$

Fig. 4 illustrates the comparison results of PAMSC-M and PAMSC on the three datasets. The figure shows that PAMSC performs better than PAMSC-M in terms of two evaluation metrics on all datasets (e.g., 70.4% vs. 69.8% in terms of classification accuracy on Yelp2020 dataset). This proves that the current mood could help PAMSC focus on the mood-congruent sentiment classes more finely, which further assists PAMSC in classifying user sentiment more accurately than PAMSC-M. Another interesting observation is



Fig. 4. The experimental result of PAMSC-M and PAMSC.

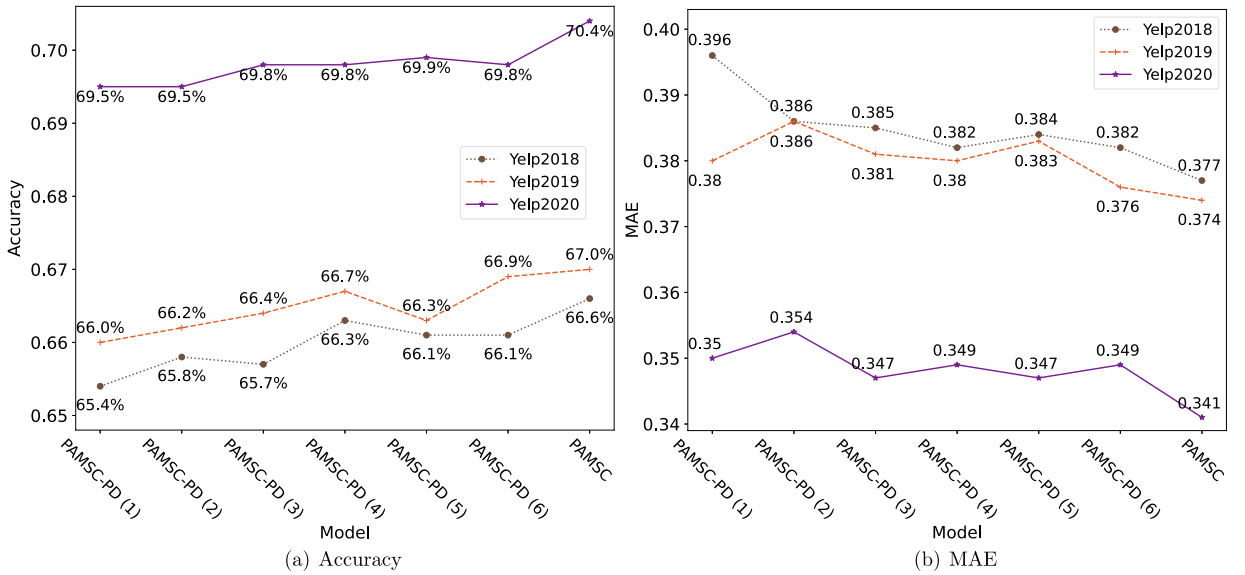


Fig. 5. The experimental results of PAMSC-PD and PAMSC.

that even without the current mood, PAMSC-M still has better classification accuracy than UserIdentifier on the three datasets (e.g., 66.2% vs. 65.6% w.r.t. accuracy on Yelp2018 dataset, see Table 3). As for the possible reason, personality may be the contributing factor, as it could influence the attention deployment of users under different moods. Next, we analyze the specific role of personality in detail.

4.7. Effect of personality

To evaluate the effect of personality on analyzing the personalized duration of the current mood (RQ3), we implement PAMSC-PD, which only adopts the proportion of positive words used in fixed blocks to model current positive/negative mood. Specifically, PAMSC-PD (k) denotes PAMSC-PD adopting the proportion of positive words used in the first k blocks of the target review. For example, the current positive mood $\mathbf{D}^{pos}$ in PAMSC-PD (k) is calculated by modifying Eq. (4):

$$\mathbf{D}^{pos} = \frac{\sum_{i=1}^k s_k^p}{\sum_{i=1}^k (s_k^p + s_k^n)} \quad (29)$$

The experimental results are shown in Fig. 5. We observe that PAMSC achieves the best performance w.r.t. two evaluation metrics on the three datasets. This demonstrates that personality could help our PAMSC model analyze the personalized duration of the current mood, which is in keeping with the earlier psychological findings [16]. Another interesting observation is that the performance of PAMSC-PD eventually improves with increasing k . For example, PAMSC-PD (6) has better performance than PAMSC-PD (1) in terms of two evaluation metrics on all datasets. The possible reason is that with the increase of k , the calculated mood becomes more stable and can represent the common moods of users so that PAMSC-PD (6) could perform better than PAMSC-PD (1).

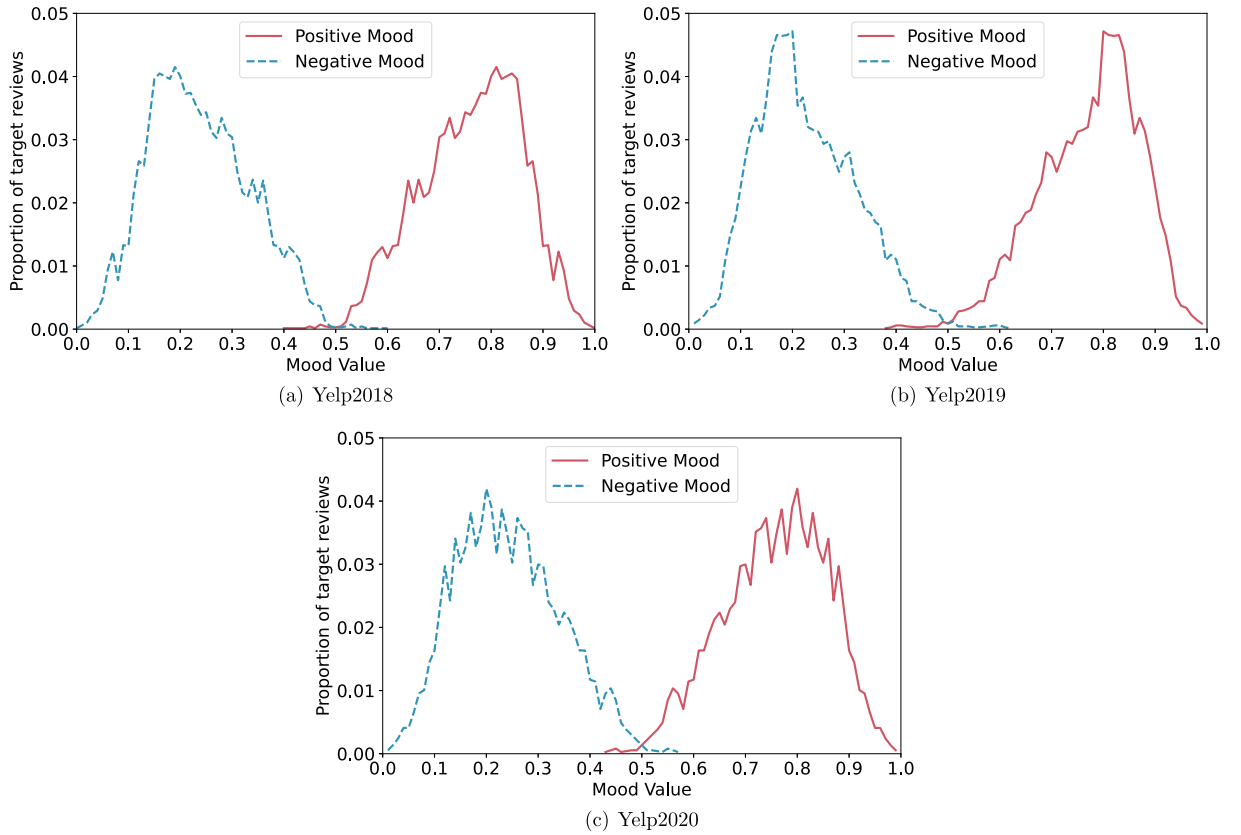


Fig. 6. The distribution of users' current positive and negative moods calculated by PAMSC. The current positive and negative moods are related to the target reviews in different test datasets.



Fig. 7. The experimental results of PAMSC-PI and PAMSC.

To intuitively observe the user's current mood calculated by our PAMSC model, we visualize the distribution of the current positive/negative moods calculated by our PAMSC model under different test sets in Fig. 6, where the value of current positive or negative mood ranges from 0 to 1. The figure shows that with the assistance of personality, the current positive moods calculated by our PAMSC model are mostly between 0.5 and 1.0, while the majority of the current negative moods are between 0.1 and 0.5. This phenomenon implies that users are usually in a relatively strong positive mood, which is a valuable discovery and could be applied to many fields. For example, online advertising could deliver more advertisements to those who have a positive mood, as users in a positive mood will note more advertisements than users who are in a negative mood [4].

To answer **RQ4**, we implement PAMSC-PI to evaluate the effect of personality for analyzing the personalized attention of users under different moods. Concretely, PAMSC-PI does not utilize the user's personality in the mood impact modeling module. It calculates A^{pos} and A^{neg} by modifying Eq. (7) and Eq. (10):

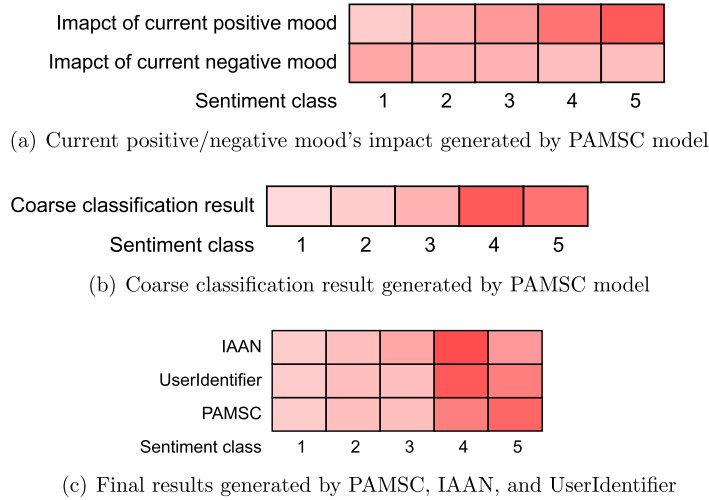


Fig. 8. Visualization of different data generated by PAMSC, IAAN, and UserIdentifier.

$$\mathbf{A}^{pos} = \text{softmax}(\mathbf{D}^{pos} * \tanh(\mathbf{Z}^{pb})) \quad (30)$$

$$\mathbf{A}^{neg} = \text{softmax}(\mathbf{D}^{neg} * \tanh(\mathbf{Z}^{nb})) \quad (31)$$

Fig. 7 shows the experimental results of PAMSC-PI and PAMSC on the three datasets. It can be seen that PAMSC-PI performs worse than PAMSC (e.g., 0.380 vs. 0.374 w.r.t. MAE on Yelp2019 dataset), which demonstrates that the user's personality could influence her or his attention toward mood-congruent information under different moods. This finding is consistent with the current studies showing that information processing is associated with both current mood and personality [36]. Hence, it is useful to adopt user personality to add personalized factors into the basic positive/negative-based attention vector. With the aid of personality, our PAMSC could analyze the personalized attention of users with different personalities toward mood-congruent information and model the personalized mood impact more finely.

4.8. Case study

In the following, we take a target review from Yelp2019 dataset as an example to intuitively display the effect of personality-assisted mood. The content is shown below. The ground truth sentiment class of this target review is “5”.

“Fast in and out. Easy location to get to. It's in a shopping center so if you need to grab some lunch, run to Whole Foods or whatever it's all in the same center. Friendly staff and again it's a quick in and out.”

For our PAMSC model, we visualize the current positive/negative mood's impact (Fig. 8(a)). The visualization of the coarse classification result and the final result are shown in Fig. 8(b) and Fig. 8(c). To better demonstrate the effect of personality-assisted mood, we also visualize the final results of the best two baseline models (i.e., IAAN and UserIdentifier) (Fig. 8(c)). The darker color in Figs. 8(a), 8(b), and 8(c) means higher attention weight or correct probability. The findings are summarized as follows:

(1) Firstly, in the mood extraction module, our PAMSC model extracts the current positive and negative moods (i.e., 0.866 and 0.134), indicating that the user is in a relatively strong positive mood.

(2) Secondly, in the mood impact modeling module, according to the theory of mood-congruent attention bias, our PAMSC model generates the impact of the current mood (Fig. 8(a)). It can be seen that our PAMSC model gives more attention to sentiment classes “4” and “5” from the view of the current positive mood. In contrast, from the perspective of the current negative mood, our PAMSC model gives more attention to sentiment class “1”.

(3) Thirdly, in the coarse classification module (Fig. 8(b)), with the consideration of the target review by global user preference, PAMSC model prejudices that the user is likely to write the target review with sentiment class of “4” or “5”, while the correct probability of sentiment class “4” is larger than that of sentiment class “5”.

(4) Finally, as seen in Fig. 8(c), after considering the impact of the current mood, PAMSC model predicts the sentiment class of the target review as “5”, which is in accordance with the ground truth sentiment class. While for other baseline models, they predict the sentiment class of the target review as class “4”, which is not correct. Actually, in the coarse classification module, our PAMSC model also prejudices that the sentiment class should be class “4” based on the target review alone. However, our PAMSC model adds the correct probability of sentiment class “5” with the help of the personalized mood impact.

Hence, the visualization of these data better shows the working mechanism of our PAMSC model, which is driven by the cognitive mechanism of mood-congruent attention bias. This mature psychology theory adds the interpretability of the whole prediction procedure and is beneficial for users to understand and trust the prediction result.

5. Discussion

In summary, the experimental results demonstrate that our PAMSC model significantly outperforms the related models w.r.t. classification accuracy on the three real-world datasets. Furthermore, compared with the related models that do not consider the user's mood, the usage of the personalized mood impact in our PAMSC model increases the interpretability of the whole prediction procedure. In addition, the user's personality proves to be useful for analyzing the duration of the user's mood and the attention toward mood-congruent information. By fully utilizing the user's personality, our PAMSC model could model the personalized mood impact in a more fine-grained way and further classify user sentiment precisely.

Even so, our PAMSC model still has some limitations:

(1) Although our PAMSC model achieves the highest classification accuracy and MAE on the three real-world datasets when compared to other baseline models, the performance of our PAMSC model w.r.t. MAE is not as good as classification accuracy. We hypothesize that the possible reason is that the mood of users is misjudged in some rare cases. The adjustment of the personalized mood impact to coarse classification result may lead to relatively large divergences between predicted sentiment classes and ground truth classes when a user's mood is misjudged, which further influences the performance of our PAMSC model in terms of MAE.

(2) Limited by the precise demand for the post time of user-generated reviews, we only conduct experiments on the Yelp dataset, which makes the experimental results lack a certain degree of generalization. To make up for this, we adopt the data before and after the COVID-19 epidemic to conduct experiments and enhance the persuasiveness of the experimental results.

6. Conclusions and future work

Most of the existing methods ignore the user's mood, which is hidden in historical reviews and could impact the user's sentiment expression. Furthermore, few of the related studies fully utilize the user's personality to model the personalized mood impact. Hence, in this work, we propose a Personality-Assisted Mood for Sentiment Classification (PAMSC) model to complete the review sentiment classification task. Concretely, we extract the user's current mood from the corresponding historical reviews and model its impact based on the cognitive mechanism of mood-congruent attention bias. In the meantime, we prejudge the sentiment of the target review according to the target review and global user preference. Finally, we fine-tune the prejudged result with the personalized mood impact to obtain the final classification result. In particular, personality plays important roles in our PAMSC model, including analyzing the duration of the current mood and the attention toward mood-congruent information. We build three real-world sentiment classification datasets (i.e., Yelp2018, Yelp2019, and Yelp2020) to evaluate the performance of our PAMSC model. The experimental results demonstrate that compared with other related models (e.g., SentiMem and UserIdentifier), our PAMSC model achieves significantly higher classification accuracy on the three real-world datasets. Moreover, the prediction process of our PAMSC model is proven to be more interpretable via visualization, which may enhance users' trust in the prediction result.

As for future work, we will continue to optimize the mood extraction module and the mood impact modeling module to increase the performance of our PAMSC model. On the one hand, we would like to analyze the relationship between mood and different personality traits to model the current mood in a more fine-grained manner, as different personality traits could affect the current mood from different perspectives [21]. On the other hand, we are interested in introducing other user information (e.g., user emotion [36]) to better accomplish the review sentiment classification task.

CRediT authorship contribution statement

Yu Ji: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Writing – original draft, Writing – review & editing. **Wen Wu:** Conceptualization, Formal analysis, Investigation, Methodology, Supervision, Writing – original draft, Writing – review & editing. **Yi Hu:** Supervision, Writing – review & editing. **Xi Chen:** Writing – review & editing. **Jiayi Chen:** Writing – review & editing. **Wenxin Hu:** Writing – review & editing. **Liang He:** Supervision, Writing – review & editing.

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Data availability

The authors do not have permission to share data.

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